

Divisão SQL



Roteiro

- Histórico
- Visão Geral
- DDL
- Exercícios



Divisão

- Considera duas relações
 - dividendo (grau $m+n$)
 - divisor (grau n)
- Grau “n”
 - Atributos do mesmo nome em ambas as relações
- Quociente
 - grau “m”
 - Atributos da relação dividendo cujos valores associam-se **TODOS** com os valores da relação divisor
 - Notação
 - **Relação 1 $\div$ Relação 2**



Divisão

- Exemplo

$$R_1$$

x	y	z
1	1	1
1	2	1
2	1	1
2	2	2
3	1	3

$$R_2a$$

z
1

$$R_1 \div R_2a$$

x	y
1	1
1	2
2	1

Divisão

- Exemplo

R_1

x	y	z
1	1	1
1	2	1
2	1	1
2	2	2
3	1	3

R_2b

y	z
1	1

$R_1 \div R_2b$

x
1
2

Divisão

- Exemplo

R_1

x	y	z
1	1	1
1	2	1
2	1	1
2	2	2
3	1	3

R_2C

y
1
2

$R_1 \div R_2C$

x	z
1	1

Divisão

- Exemplo

R_1

x	y	z
1	1	1
1	2	1
2	1	1
2	2	2
3	1	3

R_2C

y
1
2

$R_1 \div R_2C$

x	z
1	1

SQL

SQL

- 1970- Foi publicado “A Relational Model of Data Large Shared Banks” que define o modelo relacional
- 1974 – Foi publicado o SEQUEL (Structured English Query Language)
- 1986 – SQL vira ISO
 - SQL-2 (92)
 - SQL-3 (99)



Visão Geral

- Possui base pautada na álgebra relacional e no cálculo relacional de tupla
- Principais funcionalidades
 - DDL – definição de dados
 - DML – manipulação de dados
 - Restrições de integridade
 - Transações
 - Stored procedures

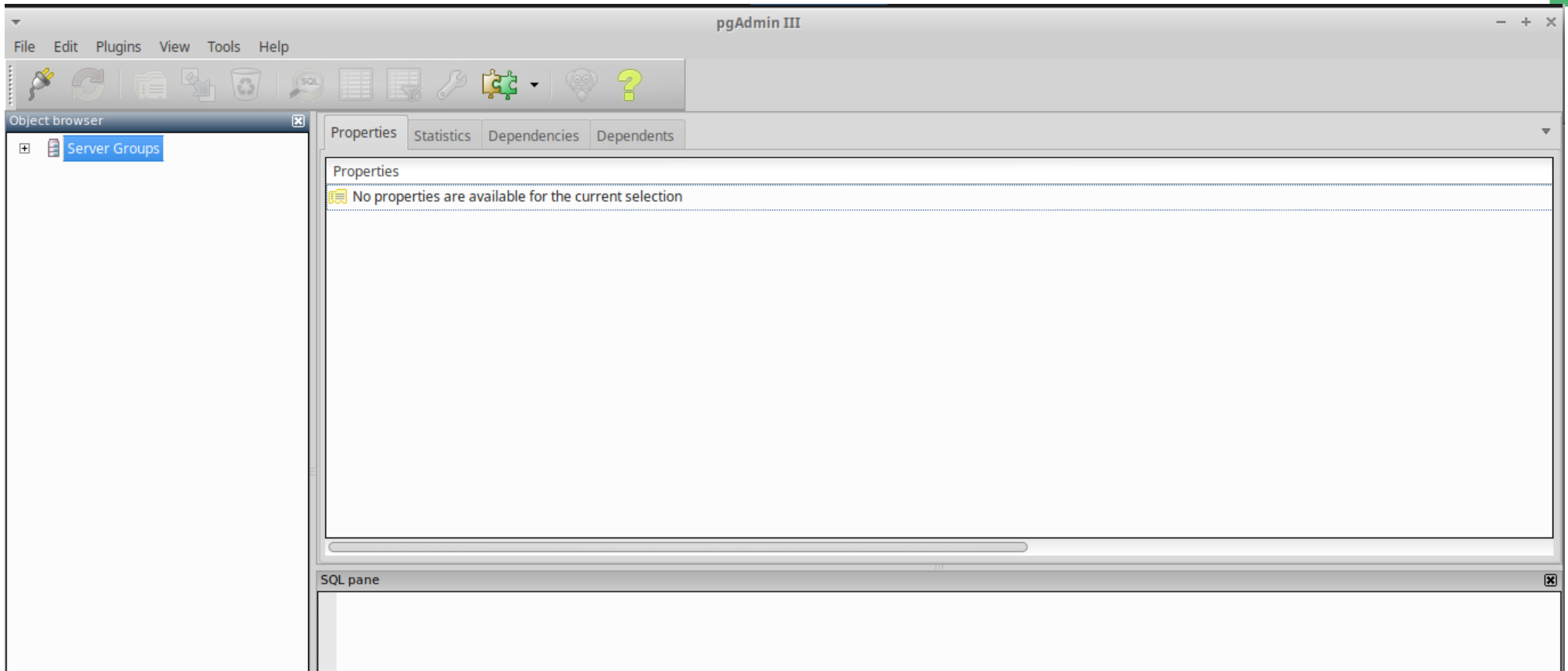


Visão Geral

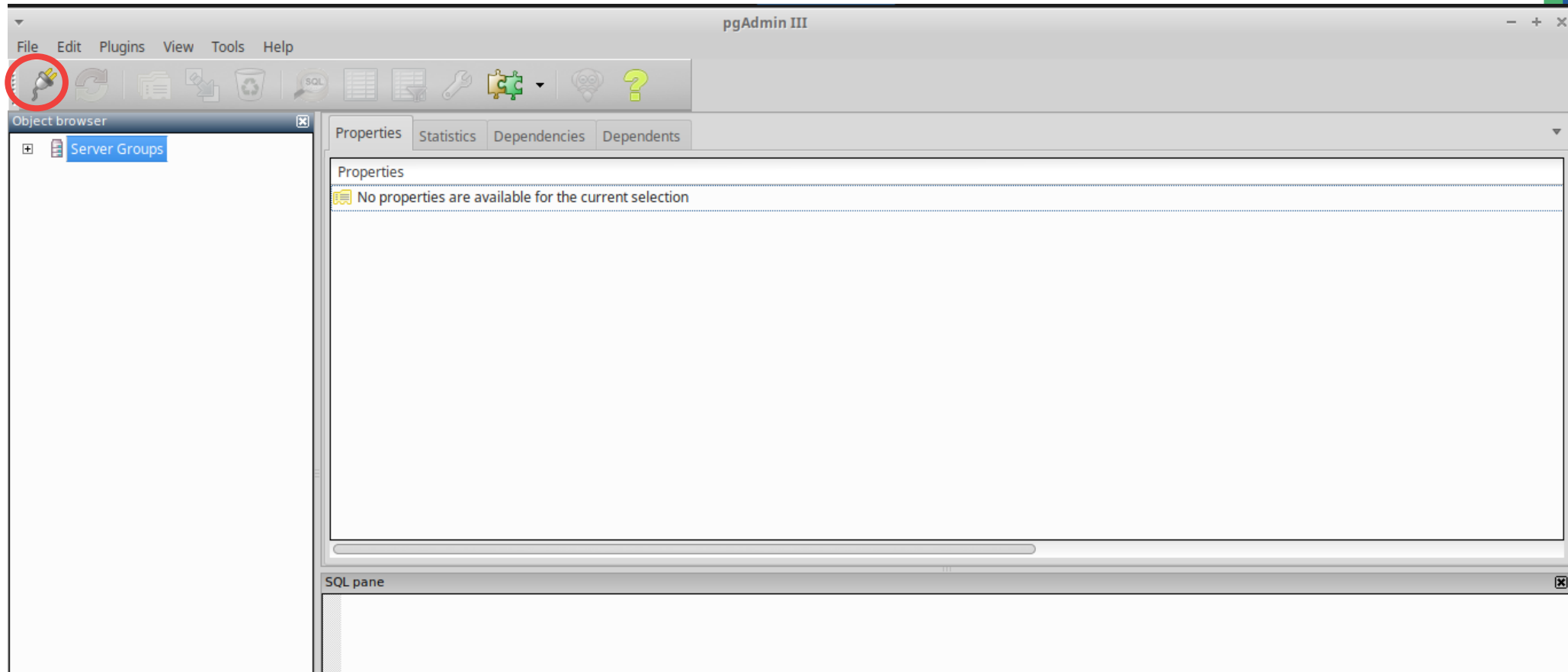
- SGBD disciplina
 - PostgreSQL
 - Interface
 - PGAdmin 3 ou 4
 - pSQL



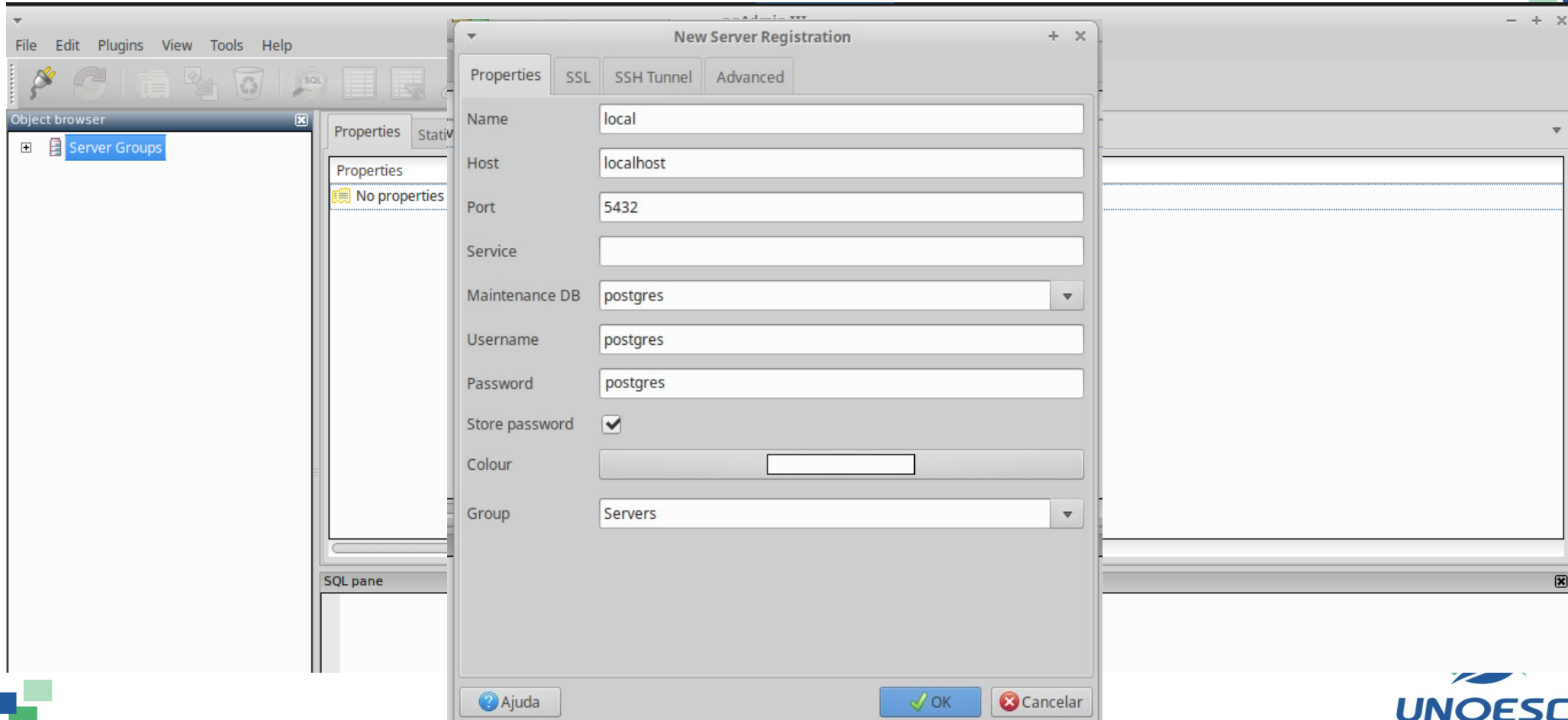
PostgreSQL + PGAdmin 3



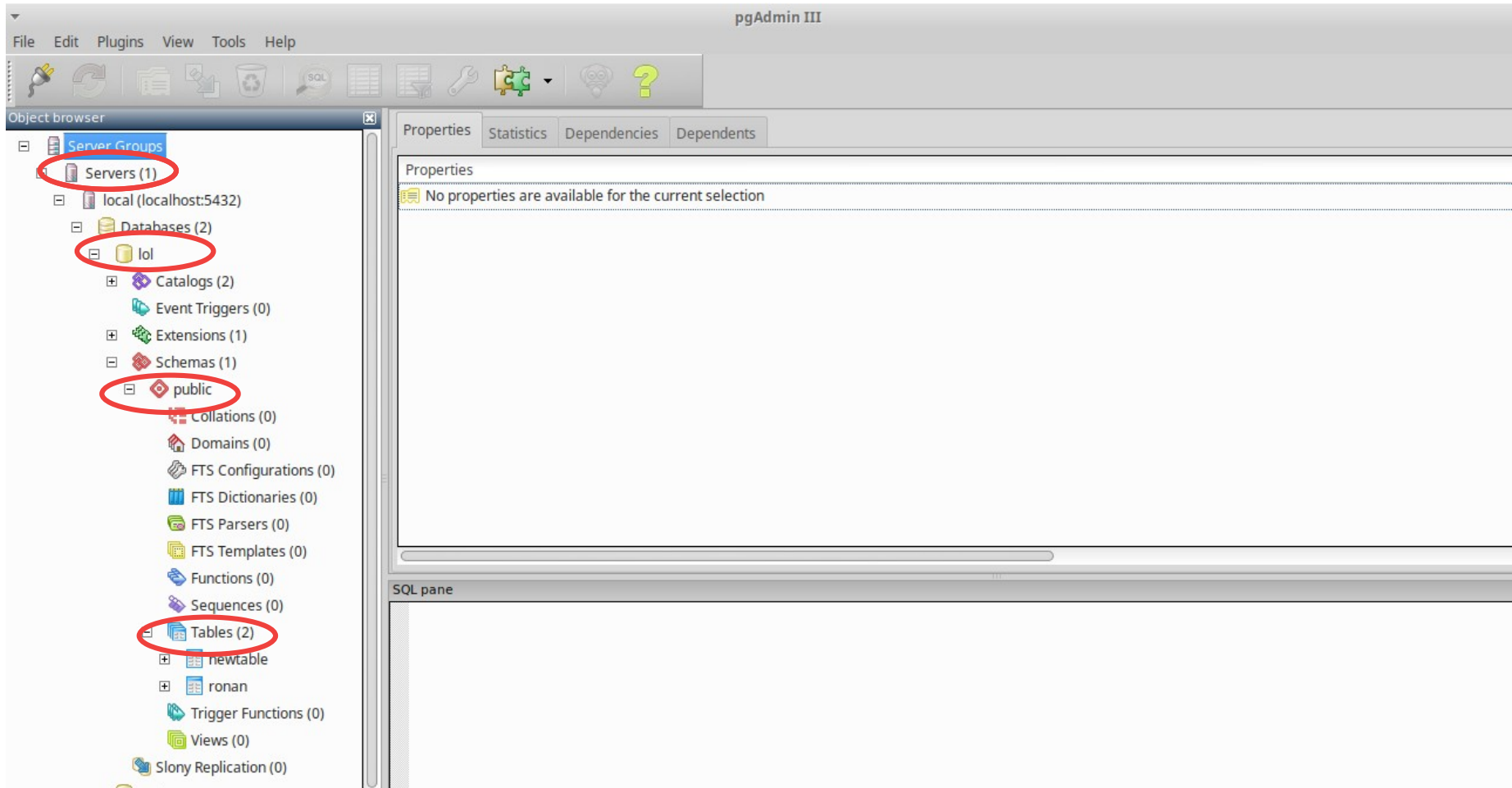
PostgreSQL + PGAdmin 3



PostgreSQL + PGAdmin 3

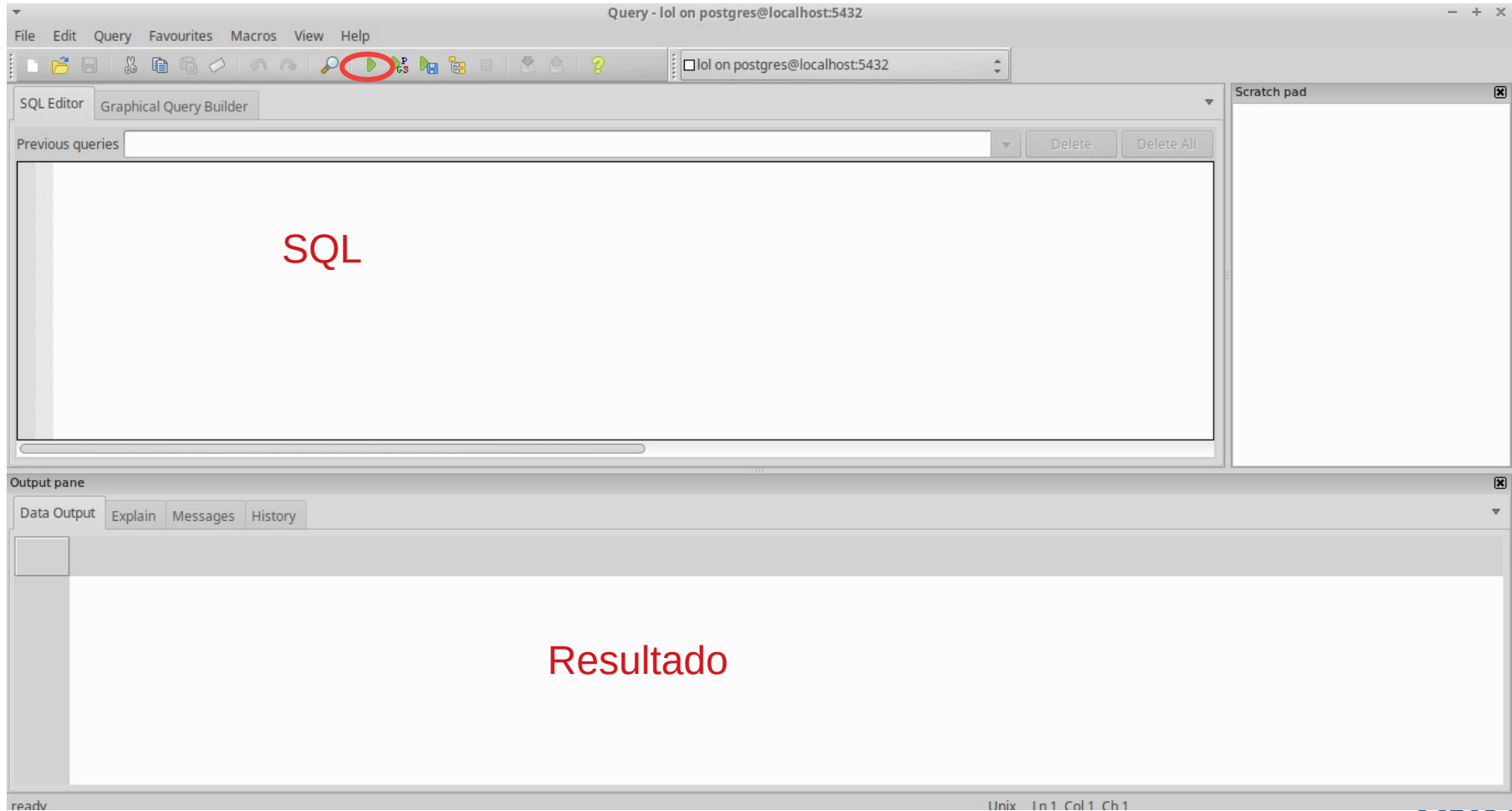


PostgreSQL + PGAdmin 3

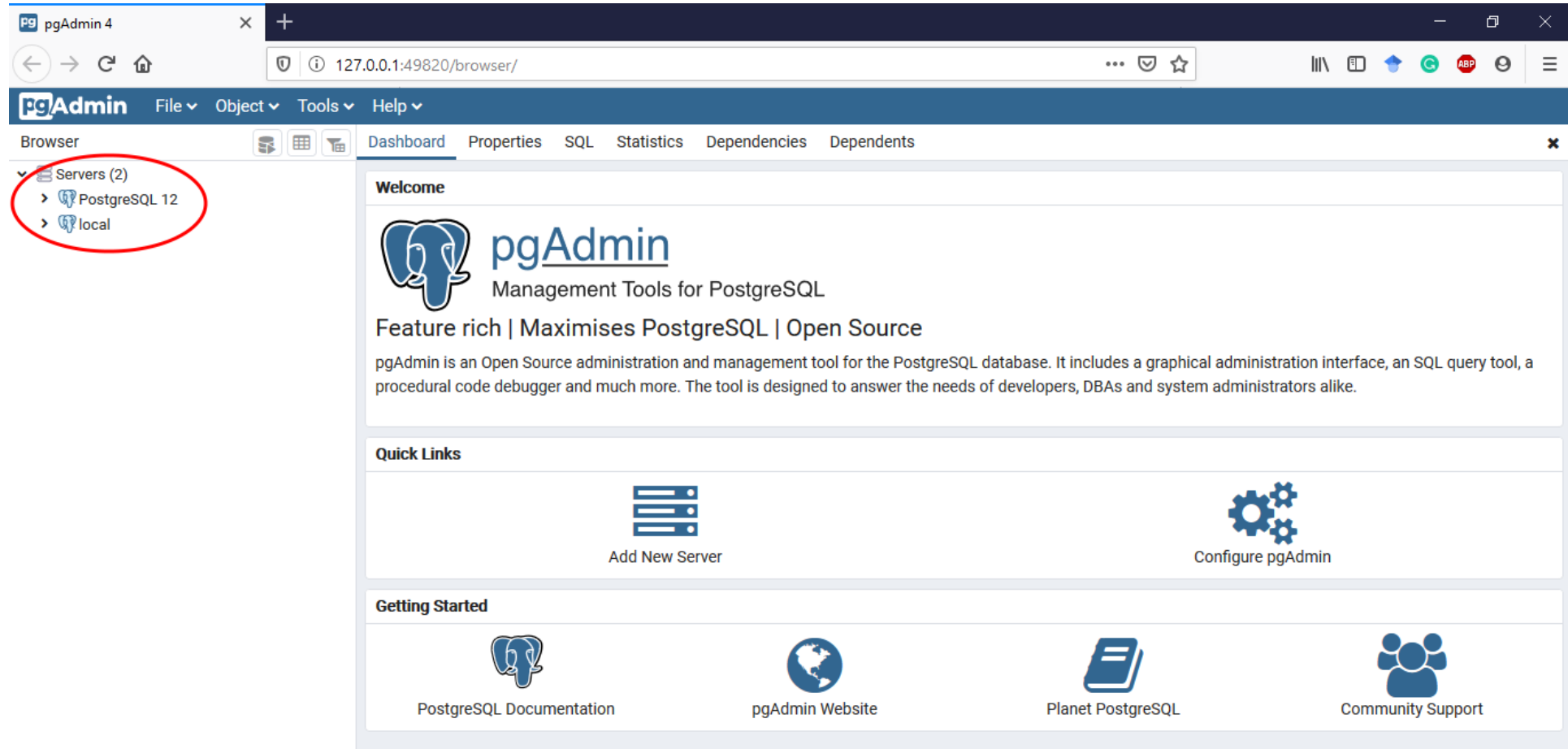




PostgreSQL + PGAdmin 3



PostgreSQL + PGAdmin 4



The screenshot shows the pgAdmin 4 web interface in a browser window. The browser's address bar displays the URL `127.0.0.1:49820/browser/`. The pgAdmin 4 logo and navigation menu (File, Object, Tools, Help) are visible at the top. Below the navigation menu, the 'Browser' tab is active, showing a tree view of servers. The tree view includes 'Servers (2)', which is expanded to show 'PostgreSQL 12' and 'local'. The 'PostgreSQL 12' server is highlighted with a red circle. The main content area displays a 'Welcome' message with the pgAdmin logo and text: 'Management Tools for PostgreSQL', 'Feature rich | Maximises PostgreSQL | Open Source', and a description of the tool. Below the welcome message, there are 'Quick Links' for 'Add New Server' and 'Configure pgAdmin'. At the bottom, there is a 'Getting Started' section with links to 'PostgreSQL Documentation', 'pgAdmin Website', 'Planet PostgreSQL', and 'Community Support'.

pgAdmin 4

127.0.0.1:49820/browser/

pgAdmin File Object Tools Help

Browser

Servers (2)

- PostgreSQL 12
- local

Welcome

 **pgAdmin**
Management Tools for PostgreSQL

Feature rich | Maximises PostgreSQL | Open Source

pgAdmin is an Open Source administration and management tool for the PostgreSQL database. It includes a graphical administration interface, an SQL query tool, a procedural code debugger and much more. The tool is designed to answer the needs of developers, DBAs and system administrators alike.

Quick Links

 Add New Server

 Configure pgAdmin

Getting Started

 PostgreSQL Documentation

 pgAdmin Website

 Planet PostgreSQL

 Community Support

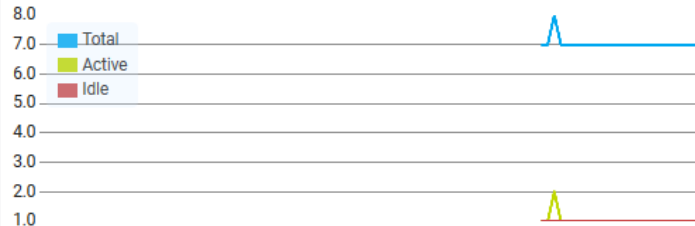
PostgreSQL + PGAdmin 4

pgAdmin File Object Tools Help

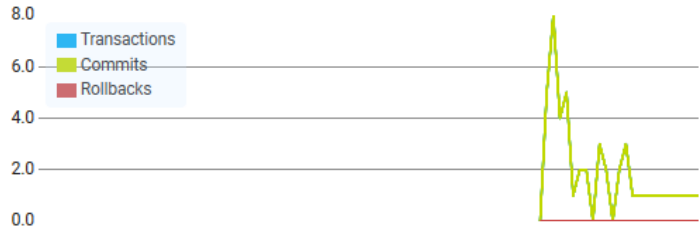
Browser Servers (2) PostgreSQL 12 Databases (1) postgres Casts Catalogs Event Triggers Extensions Foreign Data Wrappers Languages Schemas (1) public Collations Domains FTS Configurations FTS Dictionaries FTS Parsers FTS Templates Foreign Tables Functions Materialized Views Procedures Sequences Tables Trigger Functions Types

Dashboard Properties SQL Statistics Dependencies Dependents

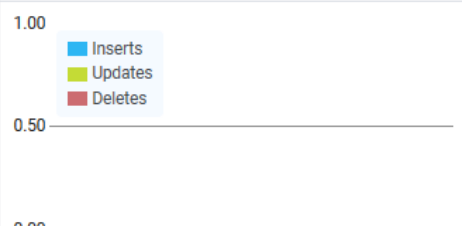
Server sessions



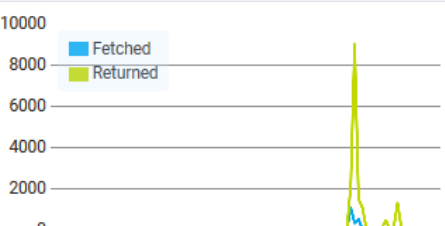
Transactions per second



Tuples in



Tuples out



Block I/O



Server activity

Sessions Locks Prepared Transactions Configuration

		PID	Database	User	Application	Client	Backend start	State	Wait event	Blocking PIDs
×	■	3576	postgres	postgres	pgAdmin 4 - DB:postgres	::1	2020-04-15 23:07:45 GMT	idle	Client: ClientRead	
○	■	4612					2020-04-15 23:04:33 GMT		Activity: CheckpointerMain	

PostgreSQL + PGAdmin 4

The screenshot displays the PGAdmin 4 web interface in a browser window. The browser's address bar shows the URL `127.0.0.1:49820/browser/#`. The PGAdmin 4 interface includes a top navigation bar with menus for File, Object, Tools, and Help. On the left, a 'Browser' pane shows a tree view of database objects: Servers (2), PostgreSQL 12, Databases (1), postgres (selected), Login/Group Roles, Tablespaces (2), and local. The main area is divided into several tabs: Dashboard, Properties, SQL (active), Statistics, Dependencies, and Dependents. Below these tabs, a toolbar contains various icons for file operations, search, and execution. The 'SQL' tab is active, showing a query editor with the text `SELECT * FROM pessoas`. Below the query editor, there are tabs for Query History, Scratch Pad, Data Output (active), Explain, Messages, and Notifications. The 'Data Output' tab is currently empty, displaying the word 'Resultados' in red. The interface is clean and professional, with a blue and white color scheme.

pgAdmin 4

127.0.0.1:49820/browser/#

pgAdmin File Object Tools Help

Browser

- Servers (2)
 - PostgreSQL 12
 - Databases (1)
 - postgres
 - Login/Group Roles
 - Tablespaces (2)
 - local

Dashboard Properties SQL Statistics Dependencies Dependents postgres/postgres@PostgreSQL 12 *

postgres/postgres@PostgreSQL 12

Query Editor Query History Scratch Pad

1 `SELECT * FROM pessoas`

SQL

Data Output Explain Messages Notifications

Resultados

psql



```
gschreiner@nasgul:~$ psql -h localhost -U postgres -d lol
psql (10.7 (Ubuntu 10.7-0ubuntu0.18.04.1))
SSL connection (protocol: TLSv1.2, cipher: ECDHE-RSA-AES256-GCM-SHA384, bits: 256, compression: off)
Type "help" for help.

lol=#
```



- Criação de um BD
 - SQL não define essa operação
 - Alguns BDs (maioria dos comerciais) possuem o comando de criação:
 - **CREATE DATABASE nome_db**
 - Remover um BD
 - **DROP DATABASE nome_db**





- Comandos para definição do esquema
 - CREATE TABLE
 - Cria uma nova tabela, com suas definições estruturais e de integridade
 - ALTER TABLE
 - Modifica as definições da tabela
 - Atributos chave não podem ser removidos
 - Atributos NOT NULL não podem ser inseridos
 - DROP TABLE
 - Remove uma tabela





- CREATE TABLE
 - CREATE TABLE tabela (
 atributo_1 tipo_1,
 [atributo_n tipo_n,]
 [PRIMARY KEY (atributo_, [atributo_n])]
 [FOREIGN KEY (nome_atributo) REFERENCES nome_tabela]
);
 - Principais tipos:
 - int, smallint, numeric(definição e tamanho), char, varchar(definição de tamanho), date, datetime, etc
 - Date → 'YYYY-MM-DD'



DDL

Name	Aliases	Description
<code>bigint</code>	<code>int8</code>	signed eight-byte integer
<code>bigserial</code>	<code>serial8</code>	autoincrementing eight-byte integer
<code>bit [(n)]</code>		fixed-length bit string
<code>bit varying [(n)]</code>	<code>varbit</code>	variable-length bit string
<code>boolean</code>	<code>bool</code>	logical Boolean (true/false)
<code>character [(n)]</code>	<code>char [(n)]</code>	fixed-length character string
<code>character varying [(n)]</code>	<code>varchar [(n)]</code>	variable-length character string
<code>date</code>		calendar date (year, month, day)
<code>double precision</code>	<code>float8</code>	double precision floating-point number (8 bytes)
<code>integer</code>	<code>int, int4</code>	signed four-byte integer
<code>numeric [(p, s)]</code>	<code>decimal [(p, s)]</code>	exact numeric of selectable precision
<code>smallint</code>	<code>int2</code>	signed two-byte integer
<code>text</code>		variable-length character string
<code>time [(p)] with time zone</code>	<code>timetz</code>	time of day, including time zone
<code>timestamp [(p)] with time zone</code>	<code>timestamptz</code>	date and time, including time zone



- CREATE TABLE
 - Exemplo
 - CREATE TABLE alunos (
matricula NUMERIC(10),
nome VARCHAR(120),
genero SMALLINT,
PRIMARY KEY (matricula)
);



- DROP TABLE
 - DROP TABLE nome_tabela;
 - Exemplo
 - DROP TABLE alunos;





- ALTER TABLE
 - ALTER TABLE nome_tabela ADD COLUMN nome_atributo tipo [, OPP nome_atributo_n tipo_n];
 - ALTER TABLE nome_tabela DROP COLUMN nome_atributo;
 - ALTER TABLE nome_tabela ALTER COLUMN nome_atributo tipo [, OPP nome_atributo_n tipo_n];
 - ALTER TABLE nome_tabela [ADD | DROP] CONSTRAINT nome_ref tipo_ref [ADD | DROP] [PRIMARY KEY ... | FOREIGN KEY ...]
 - ALTER TABLE nome_tabela RENAME COLUMN nome_atributo novo_nome_atributo;



- ALTER TABLE

- Exemplo

- ALTER TABLE Ambulatorios ADD nome VARCHAR(30)
 - ALTER TABLE Medicos DROP PRIMARY KEY
 - ALTER TABLE Pacientes DROP COLUMN doenca, DROP COLUMN cidade
 - ALTER TABLE Funcionarios ADD FOREIGN KEY(nroa) REFERENCES Ambulatorios



Exercícios

- 1) Crie um BD com nome Clinica
- 2) Crie as seguintes tabelas neste BD, considerando que os atributos sublinhados são chaves primárias e os em itálico são chaves estrangeiras:
 - Ambulatorios: nroa (int), andar (numeric(3)) (não nulo), capacidade (smallint)
 - Medicos: codm (int), nome (varchar(40) não nulo), idade (smallint não nulo), especialidade (char(20)), CPF (numeric(11) único), cidade (varchar(30)), *nroa* (int não nulo)
 - Pacientes: codp (int), nome (varchar(40) não nulo), idade (smallint não nulo), city (char(30)), CPF (numeric(11) único), doença (varchar(40) não nulo)
 - Funcionarios: codf (int), nome (varchar(40) não nulo), idade (smallint), CPF numeric(11) único, cidade (varchar(30)), salario (numeric(10)), cargo (varchar(20))
 - Consultas: codm (int), codp (int), data (date), hora (time)
- 3) Crie a coluna nroa(int) na tabela Funcionários
- 4) Remover as colunas cargo e nroa da tabela de Funcionarios
- 5) Renomeie a coluna 'city' da tabela pacientes para 'cidade'